

Architect's Directive to Cursor Dev Agent – xChat Multi-Agent Orchestration Deep Dive (Portfolio Strategy Builder Extension)

Team,

Building directly on the **xChat Polish for HNW Options Strategy Mastery** spec I delivered earlier, we are now **deepening the multi-agent orchestration layer** for the new `portfolio_strategy_build` job type. This is **not** a green-field implementation. We extend the **shipped Phase 1 xChat Hardcore** baseline (see `PLAN.md § xChat Hardcore`, `atx-multi-agent.md`, `atx-multi-agent-design-loop.mmd`, and `context-routing-multi-agent-policy.md`).

The entire flow must remain **async batch job** so the user's xChat thread stays responsive. No blocking the composer. The user receives an immediate "Job queued – correlationId: XXX" message, then a follow-up completion notification (in-thread + optional workspace toast) when the artifact is ready. This mirrors the existing `strategy_jobs` handoff to `xStrategyBuilder` but scales to **portfolio-level, multi-account, template-driven** construction.

1. Job Contract Extension (Spring `StrategyJobService` + Next BFF)

- **New jobType:** `"portfolio_strategy_build"`

- **Payload schema** (Zod-validated on Next, passed through BFF):

```
TypeScript
{
  jobId: string (uuid v7),
  correlationId: string,
  personaId: ObjectId,           // advisor persona
  templateId: "wheel" | "iron-condor" | "covered-call" | "bull-put-
spread" | "collar",
  portfolioIds: ObjectId[],      // one or more (support cross-portfolio
correlation)
  preferences: {                 // per-account or global; captured via
NL slots
    [accountId: string]: {
      riskOutlook: "conservative" | "balanced" | "aggressive",
      capitalAllocationPct: number (5-25 default conservative),
      timeHorizonDays: number,
      primaryGoal: "income" | "appreciation" | "protection",
      ivOutlook: "high" | "neutral" | "low",
      hardConstraints: string[], // e.g. ["do-not-sell-calls-on-
NVDA"]
      taxConsiderations: boolean
    }
  },
  mode: "Auto" | "Expert" | "Heavy", // maps to Grok response depth +
sub-agent count
  userWorkspaceSnapshot: { ... }      // lightweight copy of
positions/balances/watchlist at job start (for audit & reproducibility)
}
```

- **Backend storage:** Extend `strategy_jobs` collection with `jobType`, `templateId`, `preferences`, `mode`, `artifact` (final JSON), and full `turns[]` array (orchestrator + sub-agent messages).
- **Hourly cap / Redis:** Reuse existing logic (`REDIS_URL` preferred, fallback Mongo count). Conservative default: **2 portfolio_strategy_build jobs per UTC hour per user**.
- **Audit:** Every job creation, turn, and completion writes to `admin_audit_events` with trigger `"xchat-portfolio-strategy"`.

2. Multi-Agent Hierarchy (extends shipped design loop)

We follow the **exact Phase 1 orchestration pattern** documented in `atx-multi-agent-design-loop.mmd` (orchestrator → sub-agents → slots_complete → artifact synthesis). New agents are **stateless, tool-equipped Grok-4.x instances** spawned via `xAI /v1/responses` (non-streaming for batch reliability).

Core agents (all run under the **advisor** persona system prompt + custom HNWI instructions):

Agent Role	Responsibility	Tools Used (existing + new)	Mode Scaling
Orchestrator (Master)	Job intake, slot validation, sub-agent dispatch, final synthesis & artifact	RAG (options-strategy collection), workspace tools	All modes
Account Analyzer (×N)	Per-account holdings scan, correlation across portfolios, equity sizing	Portfolio/positions/watchlist tools, code interpreter	Auto: 1 summary; Heavy: deep
Strategy Leg Builder	Generate specific legs for chosen template (e.g. Wheel: CSP → CC cycle)	Option chain tools, RAG coreskills	Expert/Heavy: full Greeks
Risk & Greeks Validator	Compute delta/gamma/theta/vega, breakeven, max loss/profit, stress tests	Code interpreter (Python/NumPy via tool), RAG	Heavy only (full Monte Carlo)
Tax & Portfolio Correlator	Tax-lot matching, wash-sale flags, cross-account exposure, liquidity check	Portfolio balances + code interpreter	Expert/Heavy
Payoff Modeler	Generate ApexCharts-ready payoff surface + probability cone	Code interpreter + existing xOptions charting logic	All modes

- **Heavy mode** spawns **all sub-agents in parallel** (Spring `@Async` + `TaskExecutor` pool, or Pub/Sub if configured).
- **Expert mode** collapses to Orchestrator + Leg Builder + Risk Validator.
- **Auto mode** uses Orchestrator + single Account Analyzer + Leg Builder (fast path).

Orchestration flow (mirrors shipped `slots_complete` handoff):

1. **xChat detects template** → NL slot collector (reuse/extend `xchat/nl-workflows/`) captures preferences → `POST /api/strategy-jobs` (new endpoint or extended).
2. **Spring StrategyJobService** creates job record (`nextRunAt` immediate) → enqueues via existing poller/ShedLock.
3. **Orchestrator turn 0**: Validates inputs, spawns sub-agents via internal tool calls (orchestrator prompt includes `run_sub_agent: {role, accountId, params}`).
4. **Sub-agents execute in parallel** (each a short-lived `POST /v1/responses` with scoped tools + RAG binding to `options-strategy` + `options-coreskills`).
5. **Aggregation turn**: Orchestrator receives sub-agent outputs → synthesizes unified recommendation (per-account legs + portfolio overlay).
6. **slots_complete**: Job status → `COMPLETE` , artifact JSON saved, `correlationId` emitted back to xChat thread via existing continuation mechanism (`previous_response_id` + `store_messages`).
7. **User notification**: In-thread message + optional workspace event (reuse recommendation event pattern).

3. RAG & Tool Integration (zero new collections)

- **Primary binding**: `options-strategy` collection (already seeded from `atx-docs/rag-collection/options-strategy/` and `options-coreskills/options-coreskills.md`).
- **Prompt injection**: Each agent's system prompt appends:

text

You are operating inside The Advisor persona for HNWI clients.
Enforce conservative position sizing ($\leq 10\%$ equity default), account-type awareness (taxable vs IRA), liquidity $\geq 2x$ margin requirement.
Always cite live mid prices, IV rank, and Greeks from workspace tools.
Output must be structured JSON for artifact merging.

- **New tool stubs** (if needed): Extend existing `atxfinance-tool-stub.md` contract:
 - `getPortfolioSnapshot(portfolioIds)`
 - `analyzeOptionChain(symbol, expirationRange, strategyTemplate)`
 - `computeStrategyGreeks(legs, underlyingPrice)`
 - `generatePayoffSurface(template, legs, params)`

4. Async Decoupling & User Experience

- **Immediate xChat reply** (in `POST /api/xchat/ask` handler):

```
text

Job queued: portfolio_strategy_build (correlationId: abc-123).
Building Wheel strategy across your 3 accounts using Conservative
outlook.
This will take 30-90 seconds depending on mode. I'll notify you here
when complete.
```

- **Completion:** New xChat message pushed into the same thread (via `xchat_logs` or direct response continuation). Include:
 - Per-account summary table
 - Recommended legs with live quotes/Greeks
 - ApexCharts payoff (embedded via existing xOptions component)
 - Risk rating badge (Conservative/Balanced/Aggressive)
 - CTA buttons: “Load into xOptions”, “Open xStrategyBuilder”, “Save as preference template”
- **Error path:** Graceful fallback message + retry option; full stack trace in audit only.

5. Implementation Checklist (production-grade, reviewer-ready)

- **Spring changes:** Extend `StrategyJobService`, add `PortfolioStrategyOrchestrator` service (reuse `@Async` + existing executor), new controller endpoints if needed (BFF proxy).

- **Next BFF:** Update `bff-proxy-routes.ts` + OpenAPI parity tests.
- **xChat side:** Extend NL preflight (`xchat/nl-workflows/`) and template handler to trigger job + poll `GET /api/strategy-jobs/{jobId}` for status.
- **UI polish:** Add “Strategy Jobs” section in workspace rail (similar to existing scanners); deep-link from xChat completion.
- **Testing:**
 - Unit: sub-agent prompts + artifact schema validation.
 - Integration: full flow with seeded portfolio data + mock xAI responses.
 - Smoke: `tests/smoke/backend-http-api-parity.test.ts` extension.
- **Docs:** Update `atx-multi-agent.md`, `strategy-engine.md`, `xchat-personas.md`, `PLAN.md` (mark as shipped), `current-state-features.md`, and add new `xchat/portfolio-strategy-orchestration.md`.
- **Guardrails:** All outputs **must** respect user’s actual holdings (no hypothetical naked options). Conservative defaults enforced at every layer.

This deepened orchestration turns xChat into a **true HNWI co-pilot**: one template click → guided preferences → async multi-agent analysis across every account → production-ready strategy artifact ready for xOptions execution preview.

Cursor team: implement the **Orchestrator + one sub-agent** first (e.g. Account Analyzer) as a minimal PR. I will validate live option-chain math and risk logic against real portfolios before we expand to full parallel sub-agents.

Ready for your first diff. Let’s ship this at the same quality level as the shipped Phase 1 Hardcore.

— The Architect

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